# MZUZU UNIVERSITY

## FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

### DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

**BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY**

**ICT 4701: SOFTWARE ENGINEERING**

**END OF SEMESTER EXAMINATION**

**DATE: FEBRUARY, 2019 TIME ALLOWED: 3 HOURS**

**INSTRUCTIONS**

1. Answer **ALL** questions.
2. Marks for each question are indicated.
3. Write your answers clearly.
4. Indicate the question number for each answer.
5. This paper contains **THREE** pages. Please check

1 a. Why is software development difficult? **[2 marks]**

b. Software Engineering is more than just writing code. Comment on that statement. **[3 marks]**

c. Some Software Engineering professionals strongly recommends that developers should avoid the waterfall development approach. However there are cases in practice where the waterfall is the recommended/best approach to be followed. Describe any specific development context in which waterfall model would be superior to the iterative and evolutionary approach **[4 marks]**

d. Describe Software requirements gathering process. **[5 marks]**

e. Describe Software re-engineering. **[3 marks]**

f. Mention the project management triple constraints that are applicable to a software development project. **[3 marks]**

2. One of the software development challenges these days is on how to model complex systems.

a. State any **two** concepts that are used to model complex systems. **[2marks]**

b. Modern approaches to software engineering make use of several tools.

i. Explain factors that are considered in choosing the best tools for your system. **[6marks]**

ii. Describe both the positive and negative impact of the use of such tools in the analysis and design of your system. **[4marks]**

c. Differentiate between each of the following pair of designs

i. Function Oriented Design Vs. Object Oriented Design. **[4 marks]**

ii. Top-down Design Vs. Bottom-up Design Model. **[4 marks]**

3a. Software Testing is one of the most important activities that should be undertaken in the development of any Software.Describe**two** reasons why software should be tested.

**[4marks]**

b. Suppose you have been requested to test a complex system that has been designed using an Object-Oriented Design approach.

i. Discuss the testing strategies, and testing techniques that would be used in testing such a system. **[10marks]**

ii. Explain how you would carry out robustness testing, and load testing on this system. **[6 marks]**

4. Consider the following problem description for an Airline Reservation System:

TT Airlines runs regular flights between Ankara and 7 other cities in Turkey. There are several daily numbered flights on regular schedule. The reservation system keeps track of passengers who will be flying in specific seats on various flights., as well as people who will form the crew.

For the crew, the system needs to track what everyone does, and who supervises whom. There are three kinds of employees in the organization, pilots, hosts and ground personnel. Only the ground personnel uses the reservation system. But the potential passengers may do reservations using the system's web interface.

There is also a frequent-flier program which shows flight points to frequent passengers and they may use their points during reservation.

TT Airlines personnel sometimes fly as passengers in scheduled flights. For their reservations, the system only requests their personnel id number. Their other info is already in the system. They are given points for each flight that they participate as a crew. Then during their reservation, they may use these points against their purchases. Besides, there is a regular 50% discount for airlines personnel and their families.

The system should be able to help to perform at least the following functions:

- Creating a new flight or modifying the attributes of a flight

- Searching for a particular flight
- Cancelling a flight on a specific day
- Making or cancelling a booking on a particular flight
- Reporting names of crew and passengers (marking TT personnel separately) on a particular flight and their payment type/discounts (cash, credit card, flyer points, crew points, etc.).

a. Draw the use case diagram for the above system. **[8 marks]**

b. Create the class diagram for the above system. **[8 marks]**

c. Explain how you coulddesign such a system if you were not using object-oriented design. **[4 marks]**

5. As you continue designing and implementing the Airline reservation system described in **question4 above**, answer the following questions.

a. Explain the interface design processes of such a system **[5marks]**

b. Write a java code on how you can implement one of the classes of your choice.**[9marks]**

c. Discusssystem maintenance issues that can be applicable in this kind of a system. **[6 marks]**

**End of Question Paper**